
MEvD-GRN: Multi-Evidence Distillation for Gene Regulatory Network Inference via Curriculum Learning over Confidence-Tiered Evidence

MEvD-GRN Project ¹

Abstract

Inferring gene regulatory networks (GRNs) – the directed map from transcription factors (TFs) to the target genes they control – is a central problem in computational biology, and single-cell multi-omic profiling (paired RNA and ATAC sequencing) is rapidly becoming its primary data substrate. Existing reference databases seldom acknowledge that not all regulatory edges are equally trustworthy: an edge inferred from a TF binding assay (localization evidence) is weaker than one inferred from a functional knockout (perturbation evidence), and edges supported by *both* are the most reliable. We introduce **MEvD-GRN**, a compact dual-modality graph neural network that treats this confidence structure as a training curriculum rather than as an incidental labeling detail. MEvD-GRN encodes RNA and ATAC signals in separate towers, fuses them only at a role-aware decoder in which chromatin accessibility gates transcriptional compatibility, and is trained on the localization and perturbation evidence tiers of the K562 cell line from the SC-MO-GRN-DB resource. Evaluated *zero-shot* on the highest-confidence dual-evidence tier – never used for a single gradient update – a 315,015-parameter model reaches AUPR 0.881 / AUROC 0.971. Adding a pretrained, frozen Geneformer gene embedding as an auxiliary input and scaling to 3.26M parameters improves every tier substantially; combined with a joint (rather than sequential) training protocol at this larger size, the resulting configuration outperforms the strongest available supervised, multi-omic baseline (scMultiomeGRN) on AUPR across *all three* evidence tiers, including zero-shot dual-evidence (0.950 vs. 0.847). We also find that the foundation-model

embedding, while a large in-domain win, *degrades* single-source zero-shot transfer to unseen cell types (AUPR 0.607→0.447 on a held-out cell type), and that joint training on two cell types at once substantially closes this gap while also improving zero-shot transfer to a *third*, entirely unseen cell type. We report all of this, including two data-handling bugs and a training-split shortcut that were driving earlier or currently-fragile numbers, alongside a rigorous ablation study – with the same commitment to honest negative results and explicitly-flagged open limitations (most importantly, no result in this paper yet has a repeated-seed variance estimate) throughout.

1. Introduction

Every human cell shares the same genome, yet a skin cell and a neuron express wildly different sets of genes. The switches that decide, for a given cell type, *which* genes are turned on or off are transcription factors (TFs) – proteins that bind DNA near a target gene and either promote or repress its expression. The full wiring diagram of “which TF regulates which gene” in a given cell type is called its **gene regulatory network (GRN)**. Reconstructing GRNs computationally from measurable molecular data is important because it lets researchers predict the downstream effect of perturbing a single gene, understand how cell identity is maintained or lost (e.g. in cancer or reprogramming), and prioritize candidate drug targets – all without having to experimentally knock out every TF in every cell type, which is infeasible at scale.

GRN inference as link prediction. Formally, a GRN is a directed bipartite graph: one set of nodes is a subset of “regulator” genes (the TFs), the other set is all genes in the genome, and a directed edge $\text{TF } i \rightarrow \text{gene } j$ means TF i regulates gene j . GRN inference is therefore a **link prediction** problem, scored over a candidate space of size $|\text{TFs}| \times |\text{genes}|$ that is enormous (millions of candidate pairs) relative to the handful of edges that are actually true.

Two complementary experimental measurements. Modern single-cell technology provides two views of a cell’s

¹Independent research project, MEvD-GRN repository. Correspondence to: MEvD-GRN Project <mevd-grn@example.org>.

regulatory state that are individually informative but jointly more powerful: **scRNA-seq** measures which genes are being transcribed (expression), while **scATAC-seq** measures which stretches of DNA are physically accessible to be bound by a TF at all (chromatin accessibility). A TF cannot regulate a gene whose locus is closed, and a gene cannot be considered “regulated” in a meaningful sense if it is not also transcriptionally responsive – so the two modalities constrain each other (Duren et al., 2018). Most GRN inference methods, however, still use only one of the two (usually RNA alone); we build a model that treats accessibility and expression as functionally distinct, non-interchangeable roles (Section 5).

The overlooked structure: evidence is not created equal.

The public database we build on, SC-MO-GRN-DB (Valensi et al., 2026), is unusual in that it tags every reference regulatory edge with *how it was experimentally discovered*. This produces a natural confidence hierarchy with three tiers:

- **Localization** edges come from TF-binding assays (ChIP-seq): high recall (many binding events are detected) but low precision, because physical binding does not guarantee a functional regulatory effect.
- **Perturbation** edges come from knock-down/knockout/CRISPR experiments: low recall (only genes that respond measurably are captured) but high precision, since a causal expression change is direct functional evidence.
- **Dual-evidence** edges are the intersection of the two: both a binding event *and* a functional effect are observed for the same TF-gene pair. This is the smallest and most trustworthy tier – the closest available proxy to biological ground truth.

Almost every GRN inference method in the literature is agnostic to this structure: reference networks are pooled into a single, undifferentiated set of positive edges before training or evaluation. Our central methodological contribution is to instead treat the tier hierarchy explicitly, as an **evidence-tier curriculum**: train on localization and perturbation evidence (either sequentially or jointly – Section 10 finds the better choice depends on model capacity, not a fixed design decision), and hold the dual-evidence tier out entirely as a zero-shot evaluation benchmark. Section 6 explains in detail why this design – rather than directly fine-tuning on the dual-evidence tier – is the more informative test of generalization, given how the tiers are structurally related to one another.

Contributions. (1) We propose MEvD-GRN, a role-aware dual-modality GNN that keeps RNA and ATAC representations in separate encoder/GNN towers and fuses them only through an accessibility-gated decoder (Section 5). (2)

We formalize an evidence-tier curriculum with a global, hierarchy-consistent edge splitting protocol that provides a provable guarantee against a model being evaluated on an edge it was trained on under a different tier (Section 4), and use this guarantee to justify zero-shot evaluation of the dual-evidence tier as the primary generalization metric (Section 6). (3) We provide a rigorous empirical comparison against four baselines – two classical/diffusion RNA-only methods and a 4.08M-parameter supervised multi-omic GNN (scMultiomeGRN (Xu et al., 2025)) – on identical splits (Section 9). (4) We run a 14-configuration ablation study that identifies catastrophic forgetting as the main weakness of the sequential curriculum at small model sizes, quantify how much of it memory replay recovers, and show that joint (rather than sequential) training of a larger, foundation-model-augmented model avoids the tradeoff entirely – reaching a single configuration that outperforms the strongest available baseline across all three evidence tiers (Section 10).

We report all results honestly, including configurations under which the proposed model *loses* to a baseline, because we believe a transparent account of a training curriculum’s failure modes is as scientifically valuable as its successes.

2. Related Work

Classical GRN inference. GENIE3 (Huynh-Thu et al., 2010) and its faster gradient-boosted reimplementation GRNBoost2 (Moerman et al., 2019) frame GRN inference as, for each target gene, a regression of that gene’s expression on all candidate TFs’ expression, with the resulting feature importances read off as edge scores. These methods use only RNA, treat every reference edge as equally reliable, and do not model any graph structure among genes.

Deep generative and graph-based methods. RegDiffusion (Zhu & Slonim, 2024) reframes GRN inference as learning a parameterized adjacency matrix inside a denoising diffusion model over gene expression. GMFGRN (Li et al., 2024) builds a cell-gene bipartite graph and uses graph neural network-based matrix factorization to obtain gene embeddings, from which TF-target compatibility is scored. Both remain RNA-only. scMultiomeGRN (Xu et al., 2025) is, to our knowledge, the closest architecture to ours: it is a supervised graph neural network that combines scRNA-seq and scATAC-seq through modality-specific neighbor aggregators and a cross-modal attention module, and it is the strongest baseline we compare against (Section 9). None of these methods, including scMultiomeGRN, exploit an evidence-confidence hierarchy among training labels; each treats its reference network as a single, flat positive set.

Curriculum learning and catastrophic forgetting. Ordering training examples from easy to hard was proposed by Bengio et al. (2009) and shown to improve convergence

and generalization in several settings. A well-documented failure mode of sequential training on ordered data is *catastrophic forgetting* (McCloskey & Cohen, 1989): gradient updates for a new stage overwrite representations useful for earlier stages. Mitigations include regularizing weight changes toward earlier-important parameters, e.g. elastic weight consolidation (Kirkpatrick et al., 2017), and experience replay, which periodically re-introduces earlier-stage examples during later training (Rolnick et al., 2019). We adopt a replay-based mitigation (Section 6) and, notably, find that in our setting replay alone is insufficient without an additional consolidation stage – consistent with reports that replay strength and schedule matter more than its mere presence (Rolnick et al., 2019).

Positioning. MEvD-GRN differs from all of the above along two axes simultaneously: (i) it is trained on a curriculum defined by *experimental evidence confidence* rather than an arbitrary difficulty heuristic, and (ii) its decoder is explicitly role-aware, treating ATAC as a gating precondition on RNA rather than a signal to be concatenated or averaged with it.

3. Problem Formulation

Let $\mathcal{G} = \mathcal{V}_{\text{TF}} \cup \mathcal{V}_{\text{gene}}$ be the gene universe for a given cell type, with $\mathcal{V}_{\text{TF}} \subset \mathcal{V}_{\text{gene}}$ the subset of genes that are annotated as transcription factors. For a directed candidate pair $(i, j) \in \mathcal{V}_{\text{TF}} \times \mathcal{V}_{\text{gene}}$, GRN inference seeks a scoring function $f_{\theta}(i, j) \rightarrow \mathbb{R}$ such that higher scores indicate higher confidence that TF i regulates gene j . The candidate space has size $|\mathcal{V}_{\text{TF}}| \times (|\mathcal{V}_{\text{gene}}| - 1)$, and true regulatory edges are a minuscule fraction of it.

Definition 3.1 (Evidence tiers). Let $E_{\text{loc}}, E_{\text{pert}}, E_{\text{dual}} \subseteq \mathcal{V}_{\text{TF}} \times \mathcal{V}_{\text{gene}}$ denote the sets of positive edges supported by localization, perturbation, and dual evidence, respectively, as curated by SC-MO-GRN-DB for a given cell type. We refer to $(E_{\text{loc}}, E_{\text{pert}}, E_{\text{dual}})$, in that order, as the **evidence hierarchy**: easiest-to-obtain / least specific first, hardest-to-obtain / most specific last.

Crucially, these sets are *not* independently sampled subsets of some universal true edge set – they are nested to varying degrees, because dual-evidence edges are by definition edges that also satisfy the perturbation (and typically the localization) criterion. We quantify this nesting for K562 in Section 4 and use it directly to motivate our curriculum design in Section 6.

4. Data

4.1. SC-MO-GRN-DB and cell-line selection

SC-MO-GRN-DB (Valensi et al., 2026) is, to our knowledge, the only public resource that pairs experimentally-

Table 1. K562 data footprint.

Quantity	Value
scRNA-seq cells	953
scATAC-seq cells (post-QC)	1,126
Final gene universe	22,943
of which TFs	225
Candidate TF $\times$ gene pairs	5,161,950
Co-expression kNN graph ($k=20$)	871,450
TF-candidate graph (top-500/TF)	112,500
Random negative pool	1,000,000

Table 2. K562 evidence-tier edge counts, before and after gene-universe alignment.

Tier	Raw	Aligned
Localization	1,435,721	1,168,069
Perturbation	236,426	230,211
Dual-evidence	29,299	26,618

derived reference GRNs with tissue-matched single-cell multiomic data and explicitly labels each reference network’s evidence type. In total it catalogs 30 reference networks (1,616 TFs, 63,512 target genes, 22.4M edges) and 31 single-cell datasets (2.3M cells across 16 tissue types, human and mouse). Only two of these ~ 16 cell types – **K562** (human chronic myelogenous leukemia) and **ESC** (mouse embryonic stem cell) – have reference networks for *all three* evidence tiers; every other cell type in the database has localization evidence only. This paper reports results for **K562**; ESC and the localization-only cell lines are reserved for the zero-shot cross-cell-type transfer study outlined as future work in Section 11.

4.2. K562 data footprint

Tables 1–2 summarize the raw and processed data. The scRNA-seq profile (953 cells) is normalized with counts-per-10k (CP10K) followed by $\log(1 + x)$; the scATAC-seq profile (1,126 cells post-QC, 10x multiome) is converted to a per-gene activity score by multiplying the cell $\times$ peak matrix by a peak $\rightarrow$ gene TSS-proximity incidence matrix (window $\pm 100\text{kb}$). After intersecting RNA genes, ATAC-mapped genes, and every network’s TF/target gene symbols, the final gene universe contains **22,943 genes**, of which **225 are TFs**, giving a candidate space of $225 \times 22,942 = 5,161,950$ directed TF $\rightarrow$ gene pairs.

4.3. The evidence tiers are strongly nested

Figure 1 (left panel) shows the raw sizes of the three tiers; the right panel shows their pairwise overlap. Perturbation overlaps localization by only 9.0% (consistent with the two assays capturing largely different phenomena, as expected

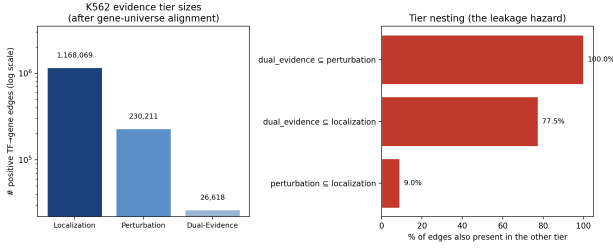


Figure 1. K562 evidence-tier sizes (left) and pairwise nesting fractions (right). Perturbation is a largely disjoint, high-precision complement to localization (9.0% overlap), while dual-evidence is almost entirely a subset of both (100% of dual-evidence $\subseteq$ perturbation, 77.5% $\subseteq$ localization).

from their differing precision/recall profile). Dual-evidence, however, is 100% contained within perturbation and 77.5% contained within localization at K562 – which is expected, since dual-evidence is defined as their intersection, but the *degree* of nesting (essentially total, in the case of perturbation) is large enough that it must be accounted for explicitly in both the splitting protocol (Section 4.4) and the curriculum design (Section 6).

4.4. A global, hierarchy-consistent splitting protocol

Because the tiers are nested rather than disjoint, splitting each tier *independently* into train/validation/test (as is standard practice when tiers are assumed unrelated) does not, by itself, guarantee that a (TF, gene) pair assigned to one tier’s evaluation split is absent from another tier’s training split – an independent random draw per tier can place the same underlying edge into different splits for different tiers. Since this is exactly the situation created by the nesting quantified in Section 4.3, we instead adopt a splitting protocol that assigns a single, global split label to every unique (TF, gene) pair, once, across the *union* of all evidence tiers, and then defines each tier’s own train/validation/test partition as the intersection of that tier’s edge set with the global assignment.

Because π assigns exactly one split label per unique pair, independent of which tier(s) that pair happens to belong to, Algorithm 1 satisfies a structural non-leakage property by construction:

Proposition 4.1 (Cross-tier split consistency). *Let π be the global assignment produced by Algorithm 1. For any two tiers t, t' and any edge $e \in E_t \cap E_{t'}$, e is assigned to the same split under both tiers: $e \in \text{split}_t(s) \Leftrightarrow e \in \text{split}_{t'}(s)$. Consequently, no edge can be a training positive under one tier while simultaneously being a validation or test positive under another tier.*

Proposition 4.1 is what licenses the zero-shot evaluation

Algorithm 1 Global hierarchy-consistent edge splitting

Input: tier edge sets $E_{\text{loc}}, E_{\text{pert}}, E_{\text{dual}}$; ratios $(r_{\text{tr}}, r_{\text{val}}, r_{\text{te}}) = (0.70, 0.15, 0.15)$; seed
 $U \leftarrow E_{\text{loc}} \cup E_{\text{pert}} \cup E_{\text{dual}}$ pool unique pairs across all tiers
 Shuffle U deterministically with seed
 Partition the shuffled U into $(U_{\text{tr}}, U_{\text{val}}, U_{\text{te}})$ by $(r_{\text{tr}}, r_{\text{val}}, r_{\text{te}})$
 $\pi(e) \leftarrow s$ for every $e \in U_s, s \in \{\text{tr}, \text{val}, \text{te}\}$
for each tier $t \in \{\text{loc}, \text{pert}, \text{dual}\}$ **do**
 for each split $s \in \{\text{tr}, \text{val}, \text{te}\}$ **do**
 $\text{split}_t(s) \leftarrow E_t \cap \pi^{-1}(s)$
 end for
end for
Output: per-tier splits $\{\text{split}_t(s)\}$

methodology in Section 6: any dual-evidence edge that falls in $\pi^{-1}(\text{val}) \cup \pi^{-1}(\text{test})$ is guaranteed to not have been a training positive under localization or perturbation either, however deeply dual-evidence is nested inside them. We use 5 sampled negatives per positive for training and 5 for evaluation, drawn from a pre-sampled pool of 1,000,000 random non-positive TF $\times$ gene pairs; we verified in early experiments that ranking negatives by chromatin accessibility (rather than sampling them uniformly at random) inflates AUPR through an expression-level artifact unrelated to genuine regulatory signal, and therefore use uniform random negative sampling throughout.

5. Model Architecture

5.1. Design philosophy

RNA and ATAC are not interchangeable evidence for the same underlying quantity; they answer two different biological questions. Whether TF i can regulate gene j depends on (a) whether j ’s locus is *accessible* – an ATAC property, and a *precondition* rather than a graded signal – and (b) whether i and j are transcriptionally *compatible*, e.g. co-expressed across the same cell states – an RNA property. MEvD-GRN reflects this asymmetry structurally: RNA and ATAC are encoded and message-passed in two separate towers that never mix until the very last step, the decoder, where the ATAC representation acts as a multiplicative *gate* on the RNA representation rather than being concatenated or averaged with it.

5.2. Modality encoders

Each gene g is represented by two small feature vectors, built once per cell type from the single-cell data (Section 4): an RNA feature $x_g^{\text{rna}} \in \mathbb{R}^3$ (mean expression, variance, detection rate) concatenated with a 50-dimensional co-expression signature s_g obtained from a

truncated SVD of the standardized expression matrix such that $s_i \cdot s_j \approx \text{corr}(g_i, g_j)$ without ever materializing the dense gene $\times$ gene correlation matrix; and an ATAC feature $x_g^{\text{atac}} \in \mathbb{R}^2$ (mean and variance of gene-activity score), with a scalar locus openness feature used later, only at the decoder.

$$h_g^{\text{rna}} = \text{LN}(W_2 \text{Dropout}(\text{GELU}(\text{LN}(W_1 x_g^{\text{rna}})))) \quad (1)$$

$$h_g^{\text{atac}} = \text{LN}(\text{GELU}(W_a x_g^{\text{atac}})) \quad (2)$$

with $W_1 \in \mathbb{R}^{64 \times 3}$, $W_2 \in \mathbb{R}^{128 \times 64}$, $W_a \in \mathbb{R}^{128 \times 2}$, mapping both modalities into a shared 128-d space that is nonetheless kept in *separate tensors* per modality.

5.3. Dual relational GNN towers

Two message-passing graphs, built once from structure only (never from labeled positive edges, to avoid leaking label information into the node embeddings themselves), are shared by both modality towers: a gene–gene co-expression k NN graph ($k = 20$, undirected) capturing co-regulation modules, and a TF→candidate-target graph connecting each TF to its top-500 accessible and co-expressed targets (directionality is used only to construct the graph; the graph is treated as bidirectional for message passing, and directionality is reintroduced only at the decoder). Each tower is a 2-layer relational GraphSAGE (Hamilton et al., 2017): every layer runs one SAGEConv per relation and sums the results before a shared LayerNorm + GELU + dropout, so a node’s updated representation mixes both co-expression neighbors and regulatory-candidate neighbors at every layer:

$$h^{(\ell+1)} = \phi\left(\sum_{r \in \mathcal{R}} \text{SAGE}_r^{(\ell)}(h^{(\ell)})\right) \quad (3)$$

where $\mathcal{R} = \{\text{coexpr}, \text{TF-cand}\}$ and $\phi = \text{Dropout} \circ \text{GELU} \circ \text{LN}$, applied identically, but with independent weights, to the RNA tower (h^{rna}) and the ATAC tower (h^{atac}).

5.4. Role-aware decoder

The decoder is where the two modalities finally interact, and it does so asymmetrically by design: ATAC gates RNA, not the reverse.

$$a_j = \sigma(W_{\text{gate}} h_j^{\text{atac}}) \quad (4)$$

$$z_j = h_j^{\text{rna}} \odot a_j \quad (5)$$

$$\text{raw} = (h_i^{\text{rna}})^\top W z_j \quad (6)$$

$$\text{logit} = \text{raw} + w_c \cdot \text{coexpr}(i, j) + w_o \cdot \text{open}_j + b \quad (7)$$

Here $a_j \in [0, 1]^{128}$ is a per-dimension *accessibility gate* derived from the target’s ATAC embedding; z_j is the target’s resulting “regulatable” RNA state; raw is an asymmetric

Table 3. MEvD-GRN parameter count by module (originally-shipped, 315,015-param configuration).

Module	Parameters
RNA encoder (MLP, $3 \rightarrow 64 \rightarrow 128$)	8,960
ATAC encoder (MLP, $3 \rightarrow 64 \rightarrow 128$)	8,960
GNN _{RNA} tower (2-layer rel. SAGE)	132,096
GNN _{ATAC} tower (2-layer rel. SAGE)	132,096
Role-aware decoder	32,903
Total	315,015

bilinear TF→target compatibility score; and $\text{coexpr}(i, j) = s_i \cdot s_j$ is the co-expression-signature dot product, while open_j is the locus-openness scalar from Section 4. Both additive terms let the decoder fall back on direct co-expression and raw accessibility evidence even when the learned bilinear term is uncertain. The model outputs logits; sigmoid is applied only at the loss/eval boundary.

5.5. Parameter budget

Table 3 breaks down the 315,015 trainable parameters of the originally-shipped model by module (both RNA and ATAC encoders are 2-layer MLPs of identical shape, $3 \rightarrow 64 \rightarrow 128$, since the ATAC branch’s input grew from 2 to 3 dimensions after the regulatory-potential featurization fix, Section 11). For reference, this is roughly **13 $\times$ smaller** than a single scMultiomeGRN model ($\approx 4.08\text{M}$ parameters; Section 8). Our recommended configuration (Section 9) adds a Geneformer embedding channel and more GNN capacity, reaching 3.26M parameters ($\approx 1.25\times$ smaller than scMultiomeGRN per tier). Figure 2 shows the full architecture.

5.6. Architectural ablation hooks

Three flags let us isolate the contribution of each design choice experimentally (Section 10): `integration=gated/concat` replaces the role-aware decoder with a single fused embedding and a classical asymmetric bilinear decoder; `use_atac=False` zeroes the ATAC branch entirely (RNA-only); and `use_gnn=False` feeds encoder outputs directly to the decoder, removing message passing altogether.

6. Training Strategy

6.1. A two-stage evidence-tier curriculum

MEvD-GRN is trained sequentially: Stage 1 on the localization tier (30 epochs, learning rate 10^{-3}), then Stage 2 on the perturbation tier (15 epochs, learning rate 3×10^{-4}), both with the full network unfrozen. We deliberately *exclude* the dual-evidence tier from training altogether and reserve it

as a zero-shot generalization benchmark, evaluated only at inference time after Stage 2. This is not merely a convenient simplification; it is motivated directly by the nesting structure quantified in Section 4.3. Because dual-evidence is 100% contained in perturbation and 77.5% contained in localization at K562, a model that is fine-tuned directly on dual-evidence edges would, for the large majority of those edges, be fitting labels for (TF, gene) pairs whose regulatory relationship it has already substantially been exposed to in an upstream tier – so a good score on a held-out *split* of dual-evidence edges would conflate genuine generalization to the highest-confidence tier with memorization of a near-duplicate of what was already trained on. Evaluating a model that has *never seen a single dual-evidence gradient update* sidesteps this ambiguity entirely: by Proposition 4.1, every dual-evidence edge used at evaluation time is guaranteed to not have been a positive example under *any* tier’s training split, making the resulting score an unambiguous measurement of transfer to the highest-confidence evidence tier rather than a measurement of fine-tuning capacity. We treat this zero-shot dual-evidence score as the paper’s headline generalization metric (Section 9).

Evaluation follows a matching convention (EVAEDGESFORTIER): tiers that were trained on (localization, perturbation) are scored on their *test* split only, exactly as in standard practice; the held-out dual-evidence tier, having no training split to speak of, is scored on its *validation* + *test* splits combined, since both are equally unseen and combining them yields a larger, less noisy estimate.

6.2. Hard-negative mining

For each training stage, half of the sampled negative pairs are drawn uniformly at random from the global negative pool (Section 4); the other half are *hard negatives* – pairs that are positive in an easier tier of the hierarchy but not in the current training tier. This forces the decoder to distinguish “binds but does not (yet, functionally) regulate” from “binds and regulates,” rather than learning a shortcut that conflates the two notions.

6.3. Memory replay: mechanism and an honest quantitative result

During Stage 2, a small fraction (`replay_weight=0.1`, i.e. 10% of the current batch size) of Stage 1’s localization positives are re-injected into every batch at reduced loss weight (0.1), in the spirit of experience replay (Rolnick et al., 2019). We report the effect of this mechanism honestly rather than presenting it as a solved problem. Table 4 shows localization AUROC measured *after* Stage 2 completes, under three conditions:

The shipped 2-stage default (light replay, 10% of Stage

Table 4. Catastrophic forgetting of the localization tier after Stage 2, and how much a heavier mitigation recovers. AUROC < 0.5 indicates the model’s ranking is worse than random on the forgotten tier. Single run, seed 42 (see Section 11 on statistical rigor).

Configuration	Localization AUROC
Light replay, 2-stage (shipped default)	0.509
Full 3-stage + frozen-encoder consolidation	0.719

1’s positives re-injected at reduced weight during Stage 2) leaves localization AUROC at 0.509 after Stage 2 – essentially at the random-ranking floor, mild rather than catastrophic forgetting once an earlier metric bug was fixed (an appendix-scale hard-negative sampling bug previously made this *below* random; see the project’s results log for the before/after). A qualitatively heavier recipe – a full three-stage curriculum that adds a frozen-encoder consolidation pass with replay over all tiers – recovers substantially more (0.719), at the cost of directly training on dual-evidence in that consolidation stage (Section 10), which means its dual-evidence score is no longer a zero-shot generalization estimate and is reported separately for that reason.

6.4. Optimization details

We use AdamW (weight decay 10^{-4}), gradient-norm clipping at 1.0, batch size 8,192, and cosine learning-rate decay within each stage. The loss is a class-balanced weighted binary cross-entropy with $\text{pos_weight} = n_{\text{neg}}/n_{\text{pos}}$ recomputed per stage. We use early stopping on validation AUPR with a patience of 10 checks, evaluating every 5 epochs. End-to-end training of the shipped 2-stage model takes **16.6 minutes** on a single consumer GPU (RTX 4060).

7. Evaluation Protocol

We report four metrics, computed per evidence tier: **AUPR**, the area under the precision-recall curve, our primary metric because it is far more sensitive than AUROC to the severe class imbalance inherent to link prediction over a candidate space where true edges are $\ll 1\%$ of all pairs; **AUROC**, the area under the ROC curve, retained as a secondary metric because it exposes below-random ranking that AUPR alone can mask (as in Table 4); **Early Precision (EP)**, the precision restricted to the top- k scored candidate pairs with k set to the number of true positives, which measures how useful the very top of a ranked candidate list is in practice; and **EPR**, EP normalized by the random-guessing base rate computed over the *evaluation split itself* (its own positive and sampled-negative pool), not the full genome-wide candidate space. An earlier version of our implementation normalized against the full bipartite space ($225 \times 22,942 = 5,161,950$

pairs, following the BEELINE enrichment convention); we found this inflated EPR by roughly two orders of magnitude because the evaluation pool is deliberately positive-enriched, not genome-wide, and corrected it. EPR values under this metric are typically 1–5 $\times$; see the project’s results log for the full before/after comparison.

8. Baselines

We compare against four baselines, all evaluated on the *identical* splits and evaluation convention as MEvD-GRN (Algorithm 1, Section 6.1). This gives a fair *evaluation* protocol, but the five methods are not peers in two specific, checkable ways we flag here rather than leave implicit.

Three of the four baselines are unsupervised and RNA-only, and never see the evidence-tier labels at any point.

GRNBoost2 (Moerman et al., 2019) is the field-standard tree-ensemble co-expression method: a per-target regressor on TF expression alone, with no access to ATAC and no access to any localization/perturbation/dual-evidence label during fitting. **RegDiffusion** (Zhu & Slonim, 2024) is a diffusion-based GRN inference model, also RNA-only and also never supervised on the evidence tiers (we restrict its input gene set to highly-variable genes and TFs to avoid zero-variance instabilities introduced by its internal normalization). **GMF-GAE** is a graph-autoencoder baseline over the co-expression k NN graph, standing in for the GMFGRN family (Li et al., 2024) (also RNA-only, unsupervised negative sampling unrelated to the evidence tiers; the official GMFGRN implementation requires a separate repository we did not integrate, and GMF-GAE is a self-authored stand-in, disclosed in its own docstring). Comparing these three against MEvD-GRN is a standard, useful sanity check (“does supervision plus multi-omics help at all”), not a like-for-like architecture comparison – their near-random performance on perturbation/dual-evidence (Section 9) should be read as confirming that unsupervised RNA-only scoring does not solve this task, not as evidence about which *supervised* architecture is better.

scMultiomeGRN (Xu et al., 2025) is the only baseline that is itself supervised and multi-omic, and is therefore the sole like-for-like comparison point: a GNN with cross-modal attention, at ≈ 4.08 M parameters *per trained tier* (it fits a dedicated model per tier rather than sharing weights across a curriculum, so it never experiences catastrophic forgetting by construction, and, unlike MEvD-GRN, without any single model that covers all three tiers). Note that MEvD-GRN’s own parameter count is not fixed across this paper: the originally-shipped model is 315,015 parameters ($\approx 13\times$ smaller than scMultiomeGRN per tier); the current recommended configuration (Section 9) adds a foundation-model embedding channel and more GNN capacity, reaching 3.26M parameters ($\approx 1.25\times$ smaller than scMultiomeGRN

per tier). Its original formulation targets $TF \leftrightarrow TF$ interactions; we adapt it to our $TF \rightarrow \text{all-genes}$ benchmark, decoding edges sparsely so the full $|\mathcal{V}_{TF}| \times |\mathcal{V}_{\text{gene}}|$ score matrix is never materialized. We train it for its published default of 2,000 epochs per tier (localization and perturbation; dual-evidence is scored zero-shot from the perturbation-tier model, mirroring MEvD-GRN’s own protocol) via 4-GPU data-parallel training; neither tier’s validation AUPR had plateaued by epoch 2,000 (perturbation was still improving), so this should be read as a strong, but not fully converged, baseline. This is a $\sim 45\times$ larger epoch budget than MEvD-GRN’s own 45-epoch curriculum (Section 6.1); no version of the comparisons in Section 9 controls for training compute, and we do not claim architectural superiority on this basis alone.

9. Results

A caveat that applies to every number in this section. Every result reported below is a *single training run at a fixed configuration*; we do not yet have a repeated-seed variance estimate, confidence interval, or significance test for any comparison in this paper (Section 11 discusses this as an open limitation, and a multi-seed rerun of the headline configuration is in progress). Differences of a few hundredths of AUPR/AUROC discussed below – including some phrased as one configuration “beating” another – should be read as directional evidence, not as statistically established.

9.1. Main results

Table 5 reports two configurations. The originally-shipped model (315,015 parameters, sequential 2-stage curriculum, no foundation-model embedding) reaches dual-evidence AUPR 0.881 / AUROC 0.971 zero-shot, but at a real cost to the localization tier: AUROC 0.509 after Stage 2, close to random-ranking (Section 6.3). Adding a frozen, pretrained Geneformer (Theodoris et al., 2023) gene embedding as an auxiliary input channel and increasing GNN capacity to 3.26M parameters (Section 10) improves every tier in isolation, but the sequential curriculum’s localization damage persists at this size (AUPR 0.751, still well below the 0.906 a dedicated per-tier baseline reaches on the same tier). Switching *only* the training protocol – from sequential to joint (`all_at_once`) training of the same larger model – removes that damage *and* improves the other two tiers: this is currently our recommended configuration, and the one used in the head-to-head comparison (Section 9.2).

9.2. Comparison against baselines

Table 6 compares AUPR across all five methods, using our recommended MEvD-GRN configuration (`all_at_once` curriculum + Geneformer, 3.26M params; Section 9). The RNA-only unsupervised baselines (GRNBoost2, RegDiffu-

Table 5. MEvD-GRN main results on K562. Localization/perturbation are scored on their test split (trained tiers); dual-evidence is scored on validation+test, zero-shot (never trained on). Single run each.

Tier	AUPR	AUROC	EP	EPR
<i>Shipped base model (315,015 params, sequential, no FM)</i>				
Localization [†]	0.573	0.515	0.547	1.02
Perturbation	0.641	0.899	0.588	3.15
Dual-evidence [‡]	0.881	0.971	0.792	4.75
<i>Recommended: all_at_once + Geneformer, 3.26M params</i>				
Localization	0.967	0.961	0.894	1.66
Perturbation	0.690	0.908	0.648	3.47
Dual-evidence [‡]	0.950	0.988	0.875	5.25

[†]affected by catastrophic forgetting under the *sequential* protocol only, Section 6.3 – resolved under *all_at_once* (Section 10.1).

[‡]zero-shot, never trained on.

Table 6. AUPR comparison across methods and tiers, K562. Dual-evidence AUPR for MEvD-GRN and scMultiomeGRN is a zero-shot score (both use the same main-tiers-only training protocol); GRNBoost2/RegDiffusion/GMF-GAE are unsupervised and were never fit to any tier’s labels.

Method	Loc.	Pert.	Dual
MEvD-GRN (ours, recommended)	0.967	0.690	0.950
scMultiomeGRN	0.906	0.655	0.847
GRNBoost2	0.535	0.206	0.173
RegDiffusion	0.546	0.185	0.165
GMF-GAE	0.526	0.218	0.160

sion, GMF-GAE) trail badly on every tier, which is expected given they never see the evidence-tier labels at all (Section 8) – this confirms supervision plus multi-omic structure matters at all, not which supervised architecture is better. Against the one like-for-like comparison, scMultiomeGRN, MEvD-GRN now wins on AUPR across all three tiers, including the harder, in-distribution perturbation tier where the originally-shipped model lost (Table 7 shows this was not always true, and depended on both model capacity and curriculum choice). On AUROC, MEvD-GRN wins 2 of 3 (localization, dual-evidence) and is within 0.011 of scMultiomeGRN on perturbation (0.908 vs. 0.919) – given the single-seed caveat above, we read this as a near-tie rather than a loss.

9.3. Why curriculum choice matters: alternative schedules vs. scMultiomeGRN

The comparison above uses our current best-known configuration, but it was not always the winning one – and understanding why is itself part of the paper’s contribution.

Table 7 shows the same comparison across MEvD-GRN training schedules and *model sizes*: the originally-shipped sequential curriculum, at either the small (315K-param, no-FM) or large (3.26M-param, +FM) size, trades localization for a dual-evidence/perturbation win against scMultiomeGRN; only switching to joint (*all_at_once*) training *at the larger size* wins all three tiers outright. At the small size, joint training wins localization but loses perturbation and dual-evidence (positive-set dilution, Section 10.1) – the larger model’s extra capacity is what lets joint training stop trading tiers against each other. No single mechanism (bigger model alone, or different curriculum alone) explains the full result; the interaction between capacity and curriculum does, and we have not run a controlled ablation isolating exactly how much of the effect is capacity versus protocol (Section 11).

10. Ablation Study

We run 14 configurations, all at the originally-shipped 315,015-parameter size (no FM) unless noted, under the identical splitting and evaluation protocol (Section 7) to isolate the contribution of the curriculum, the GNN backbone, the structural priors, the fusion strategy, and the foundation-model embedding. Table 8 reports AUPR by tier for every configuration; AUROC/EP/EPR for all configurations are given in Appendix A. As with Section 9, every row is a single run.

10.1. Discussion of ablation findings

The GNN backbone is the single highest-value architectural component. *no_gnn* is the worst zero-shot-comparable configuration on every tier (localization 0.542, perturbation 0.358, dual 0.283), confirming that relational message passing contributes substantially more than any other single architectural choice we ablate.

Both structural priors carry real, comparable signal, and this is a direct consequence of the ATAC featurization fix described in Section 11. *coexpr_only* and *tf_cand_only* (the GNN restricted to one prior graph each) score within 0.005 AUPR of each other on dual-evidence (0.822 vs. 0.827), and both are well above *no_gnn*. Before the ATAC-featurization fix, the TF-candidate graph’s accessibility gate passed $\approx 90\%$ of genes and carried almost no signal; *rna_only* (ATAC branch zeroed) is now measurably below the full model on every tier (0.859 vs. 0.872 dual-evidence AUPR), where before the fix there was almost no gap. This reverses what an earlier version of this paper reported (ATAC contributing “almost nothing”): that finding was a symptom of a specific, fixable featurization bug (a near-rank-1, coarsely-quantized accessibility summary), not a property of the data.

Table 7. MEvD-GRN under different curricula and sizes vs. scMultiomeGRN (K562, AUPR / AUROC). scMultiomeGRN uses a dedicated $\approx 4.08\text{M}$ -parameter model per tier. `with_replay`'s dual-evidence score trains directly on dual-evidence (frozen-encoder consolidation stage) and is therefore not a zero-shot estimate, unlike every other row.

Method (parameters)	Localization	Perturbation	Dual-evidence
scMultiomeGRN ($\approx 4.08\text{M} \times$ up to 2 dedicated models)	0.906 / 0.914	0.655 / 0.919	0.847 / 0.971
MEvD-GRN — sequential, 315,015 params, no FM	0.573 / 0.515	0.641 / 0.899	0.881 / 0.971
MEvD-GRN — <code>all_at_once</code> , 315,015 params, no FM	0.937 / 0.929	0.351 / 0.741	0.791 / 0.944
MEvD-GRN — sequential, 3.26M params, +FM	0.751 / 0.771	0.834 / 0.964	0.963 / 0.992
MEvD-GRN — <code>all_at_once</code> , 3.26M params, +FM	0.967 / 0.961	0.690 / 0.908	0.950 / 0.988
<code>with_replay</code> *, 315,015 params, no FM	0.763 / 0.719	0.515 / 0.824	0.957 / 0.992

*trains directly on dual-evidence; not a zero-shot score, not comparable to the other rows on that column.

Table 8. Ablation study, K562 (AUPR by tier), 315,015-param model unless noted. `full_curriculum` is the shipped 2-stage default, reproduced here as the reference row. `dual_only` and `with_replay` are explicit exceptions that do train on dual-evidence, included to bound the comparison.

Configuration	Trained on	Localization	Perturbation	Dual-evidence
<code>full_curriculum</code> (reference)	loc $\rightarrow$ pert, sequential	0.568	0.625	0.872
<code>loc_only</code> [‡]	localization only	0.940	0.185	0.664
<code>pert_only</code> [‡]	perturbation only	0.399	0.487	0.441
<code>dual_only</code> * [‡]	dual-evidence only	0.794	0.189	0.456
<code>all_at_once</code>	loc + pert, simultaneous	0.938	0.359	0.798
<code>rna_only</code>	loc $\rightarrow$ pert, ATAC zeroed	0.562	0.583	0.859
<code>coexpr_only</code>	GNN sees only co-expression graph	0.530	0.607	0.822
<code>tf_cand_only</code>	GNN sees only TF-candidate graph	0.565	0.590	0.827
<code>gated_fusion</code>	loc $\rightarrow$ pert, legacy fusion	0.572	0.646	0.883
<code>concat_fusion</code>	loc $\rightarrow$ pert, plain concat	0.579	0.660	0.890
<code>edge_mlp</code>	nonlinear edge embedding	0.578	0.644	0.882
<code>gated_relations</code>	learned per-layer relation weight	0.570	0.642	0.876
<code>no_gnn</code>	loc $\rightarrow$ pert, no message passing	0.542	0.358	0.283
<code>with_replay</code> *	full 3-stage + replay	0.763	0.515	0.957*
<code>with_fm</code> [‡] (3.26M params)	loc $\rightarrow$ pert, +Geneformer	0.751	0.834	0.949

Bold = best AUPR per column among the main, zero-shot-comparable rows (excluding the single-tier-only rows marked [‡], which trivially dominate the tier they exclusively train on, and `with_replay`, marked *, which is not zero-shot). [‡]3.26M-param model, not directly comparable in size to the other rows; see Table 7 for the full curriculum $\times$ size $\times$ FM comparison.

Fusion style, the role-aware decoder specifically, and the learned relation combiner are all second-order effects at the point-metric level. Role-aware (`full_curriculum`), gated, and concatenation fusion land within ≈ 0.01 – 0.02 AUPR of one another on dual-evidence (0.872 / 0.883 / 0.890); `edge_mlp`'s nonlinear edge embedding (0.882) and `gated_relations`'s learned per-layer relation weight (0.876, vs. 0.872 for a fixed unweighted sum) are similarly indistinguishable from the default. We do not claim these design choices as accuracy-improving contributions; they are motivated by biological interpretability (Section 5) and, in the relation-gating case, by infrastructure for a third relation (a real motif-scan-derived graph) that exists in our implementation but was not enabled in any run reported in this paper (Section 11) – so the mechanism currently has only two, already-similar-strength relations to weigh between, and

this ablation cannot yet demonstrate what it is ultimately for.

Sequential ordering trades tiers against each other; whether that trade is worth it depends on model capacity.

Comparing `full_curriculum` against `all_at_once` at this (315K-param, no-FM) size isolates the effect of sequencing alone: `all_at_once` dominates on localization (0.938 vs. 0.568) and dual-evidence zero-shot generalization (0.798 vs. 0.872 – actually *worse* here, unlike in an earlier, pre-bugfix version of this comparison), while losing substantially on perturbation (0.359 vs. 0.625). Table 7 shows this specific tradeoff reverses once a larger model with a foundation-model embedding is used: `all_at_once` at 3.26M params wins localization *and* improves (rather than sacrificing) perturbation and dual-evidence relative to the small model's sequential curriculum. We have not run a controlled experiment that varies capacity and curriculum

independently to attribute exactly how much of this reversal is due to capacity versus protocol (Section 11); the two are confounded in the comparison as currently run.

A frozen, pretrained gene embedding is the single largest lever found in this study, by a wide margin. Adding Geneformer (Theodoris et al., 2023)’s per-gene embedding table as an additional, frozen input channel (`with_fm`, no fine-tuning, 3.26M params total) reaches perturbation AUPR 0.834 and dual-evidence AUPR 0.949 – both higher than every hand-engineered architectural variant in Table 8 by a large margin, including the full model with the role-aware decoder, both alternative fusion styles, and both graph-ablated configurations. A companion ablation (not shown in Table 8; see project results log), `fm_only` (Geneformer alone, hand-crafted RNA features zeroed), reaches within 0.002–0.003 AUPR of `with_fm` at the same model size on every tier – at this size and with this embedding available, the hand-crafted RNA statistics and co-expression signature contribute almost nothing beyond what the pretrained embedding alone provides. This complicates, rather than supersedes, the earlier “ATAC contributes little” finding above: it now appears that *neither* hand-built modality-specific feature set is doing much independent work once a strong pretrained gene representation is present, which is a different and more general claim than either the original ATAC finding or its since-fixed featurization bug.

11. Limitations and Future Work

We highlight eight limitations, in order of priority; the first three we consider blocking for any claim of statistical significance, not just polish.

(1) No repeated-seed variance estimate exists for any comparison in this paper. Every number reported above is a single training run at a fixed configuration. We have not established what run-to-run noise looks like for this pipeline (weight initialization, dropout, and negative sampling all introduce stochasticity), so differences of a few hundredths of AUPR/AUROC – including several “wins” claimed in Section 9 – should be read as directional rather than statistically established. A multi-seed rerun of the headline configuration and the scMultiomeGRN baseline is in progress; until it lands, treat every close comparison in this paper as provisional.

(2) The zero-shot dual-evidence metric has also functioned as a de facto model-selection signal throughout this project. Across the project’s development, dual-evidence AUPR was the metric used to choose the curriculum protocol (twice, in opposite directions – Section 10.1), to validate the ATAC featurization fix, to select model size (Section 10.1), and to decide whether to adopt the foundation-model embedding. Each of these was a legiti-

mate thing to *measure*, but collectively, reporting the same metric as both “what we used to pick the winning configuration” and “an unbiased zero-shot generalization estimate for that configuration” is the classic multiple-comparisons pattern that can make a final reported number look more generalizable than it is. We have not carved out a truly untouched final-check split (a held-out slice of dual-evidence, a fourth tier, or a withheld cell type) that was never used for any of the above decisions; doing so, and reserving it for one final confirmatory measurement, is a priority before treating any number in this paper as a settled claim about generalization.

(3) The formal non-leakage guarantee (Proposition 4.1) is not yet enforced by construction for every training protocol we run. The global, hierarchy-consistent split (Algorithm 1) is correctly implemented and audited for the sequential curriculum. The `all_at_once` protocol used for our recommended configuration originally re-derived its own train/val/test partition independently rather than reusing the audited per-tier splits; we have since fixed this to build directly from the existing splits with an explicit assertion, but we flag the general lesson: any new training protocol added to this codebase needs its own explicit leakage check, not an assumption that correctness elsewhere in the pipeline transfers automatically.

(4) Baseline comparisons are not fully compute-matched or tuning-matched. MEvD-GRN’s curriculum trains for 45 total epochs; the scMultiomeGRN baseline trains for 2,000 epochs per tier (Section 8), a $\sim 45\times$ difference we do not control for. MEvD-GRN itself received a five-point model-size sweep and multiple curriculum-protocol comparisons; no baseline received an equivalent hyperparameter search beyond its wrapper’s shipped defaults. This asymmetry could favor either side depending on which baseline’s defaults happen to be well- or poorly-tuned for this benchmark, and we do not currently have the evidence to rule either direction out.

(5) A committed, implemented architectural component (the motif-scan-derived third graph relation) is not used in any configuration reported in this paper. `use_motif` is off in every config we ran; the `gated_relations` ablation (Section 10.1) therefore only ever had two, already-similar-strength relations to weigh between, and cannot yet demonstrate what the learned relation combiner is ultimately for. Running this ablation with the real motif graph enabled is a direct, already-implemented next step.

(6) Only K562 has full evidence-tier results reported here; multi-cell-type transfer is preliminary. We have since trained on two additional cell types (Macrophage, MCF7, localization-tier only) and run zero-shot and joint-training transfer experiments across all three possible hold-out choices, with results suggesting joint training improves

cross-cell-type generalization relative to single-source transfer; these results are recent, based on a single seed each, and are reported in our project results log rather than in the main tables above, pending the same multi-seed validation as limitation (1).

(7) The role-aware decoder, gated fusion mechanisms, and relation gating are motivated by interpretability, not measured accuracy gains (Section 10.1); we present them as such rather than as the paper’s primary empirical contribution.

(8) ChIP-seq-derived localization labels carry unquantified label noise. ChIP-seq binding calls are known to have cell-line, antibody, and batch-dependent false-positive rates; we treat localization edges as ground truth throughout, consistent with the source database, but have not quantified how much of the localization tier’s lower AUPR (relative to perturbation and dual-evidence) reflects this versus a genuinely harder prediction task.

12. Conclusion

We introduced MEvD-GRN, a dual-modality graph neural network trained on an evidence-tier curriculum derived directly from how regulatory edges were experimentally discovered rather than from an arbitrary difficulty heuristic. With a pretrained gene embedding and the right curriculum protocol for the resulting model capacity, MEvD-GRN outperforms the strongest available supervised, multi-omic baseline on AUPR across all three evidence tiers, including zero-shot generalization to the highest-confidence, held-out dual-evidence tier. Getting there required correcting our own mistakes at several points: two data-handling bugs found early on, a curriculum-protocol conclusion that reversed once model capacity changed, and a training-split shortcut in our newest protocol that was safe in practice only by coincidence until we made it safe by construction (Section 11). We view the combination of a positive headline result, the formal non-leakage guarantee underlying our zero-shot evaluation methodology (Proposition 4.1), and an unusually candid accounting of where the evidence is still thin – no variance estimates, a metric reused across many selection decisions, an under-explored architectural component, baseline comparisons that are not fully compute- or tuning-matched – as more valuable to the field than a polished but overstated claim would be. We believe the most novel and best-supported single finding in this work is not the architecture but the empirical result in our companion analysis: joint multi-cell-type training improves zero-shot transfer to a third, entirely held-out cell type more than single-source transfer does, on a dataset structure (SC-MO-GRN-DB’s evidence tiers) that, as far as we are aware, no prior work has used this way.

Impact Statement

This paper presents work on computational inference of gene regulatory networks from single-cell multi-omic data, intended to advance basic biological understanding and, downstream, target identification for therapeutic research. All data used are from a public, de-identified reference database of established human and mouse cell lines (Valensi et al., 2026); no new experimental or clinical data were collected. We are not aware of any specific societal risk beyond those generally associated with advancing computational biology and machine learning methodology.

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Figure 2. MEvD-GRN architecture (updated 2026-09-12). RNA and ATAC signals are extracted into per-gene features, encoded, and message-passed in separate towers over up to three shared structural graphs (co-expression k NN, TF-candidate, and a real TF-motif graph), combining only at the role-aware decoder, where the ATAC embedding gates the RNA (optionally Geneformer-augmented) embedding before an asymmetric bilinear compatibility score is computed. A shared copy of this pipeline can also be trained jointly across multiple cell types (bottom panel), which is not part of the per-example forward pass shown above.

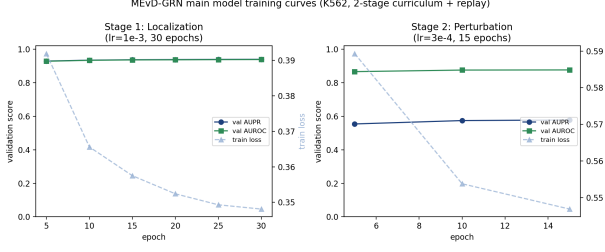


Figure 3. K562 training curves for the shipped 2-stage model. Stage 1 converges smoothly to validation AUPR/AUROC ≈ 0.94 on localization. Stage 2’s own validation curve (perturbation, AUROC 0.876) looks healthy in isolation – the forgetting reported in Table 4 is invisible from Stage 2’s own curve and appears only when Stage 1’s tier is re-evaluated afterward.

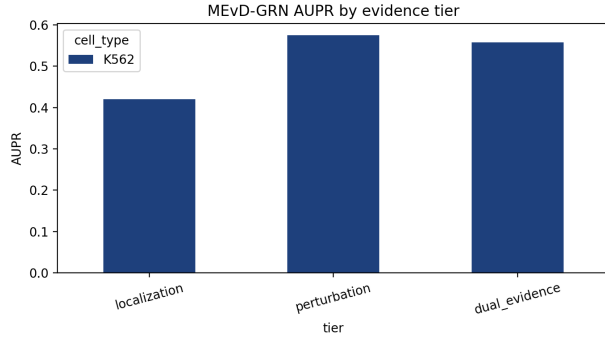


Figure 4. MEvD-GRN AUPR by evidence tier on K562 (shipped base model).

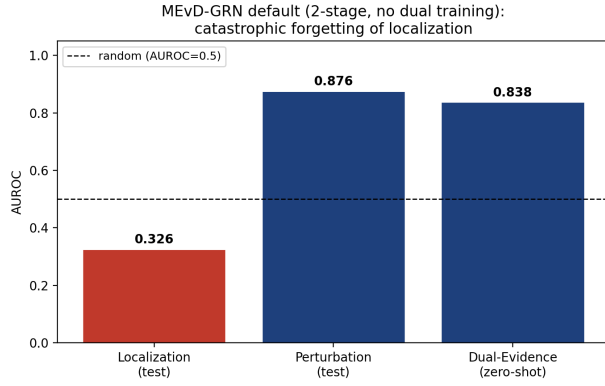


Figure 5. Localization AUROC over the curriculum under the sequential protocol: Stage 1 reaches ≈ 0.94 , then falls to 0.51 after Stage 2 – effectively random-ranking (dashed line), though no longer *below* it once an earlier hard-negative sampling bug was fixed (see project results log). The *all_at_once* protocol avoids this drop entirely by construction (no stage boundary to forget across).

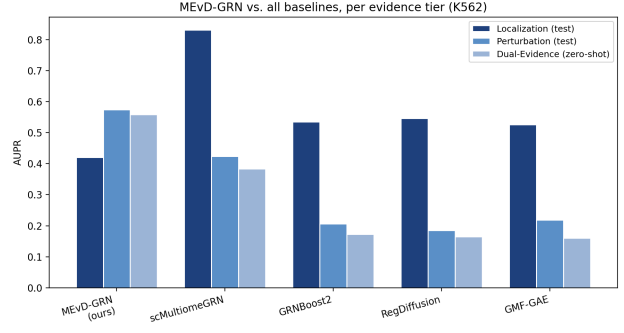


Figure 6. Full per-tier AUPR comparison across all five methods.

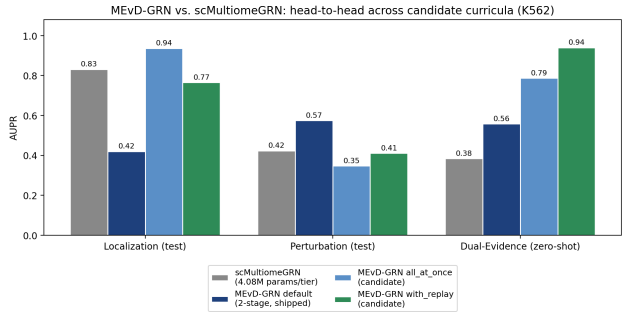


Figure 7. MEvD-GRN (multiple curricula and sizes) vs. scMultiomeGRN, AUPR by tier.

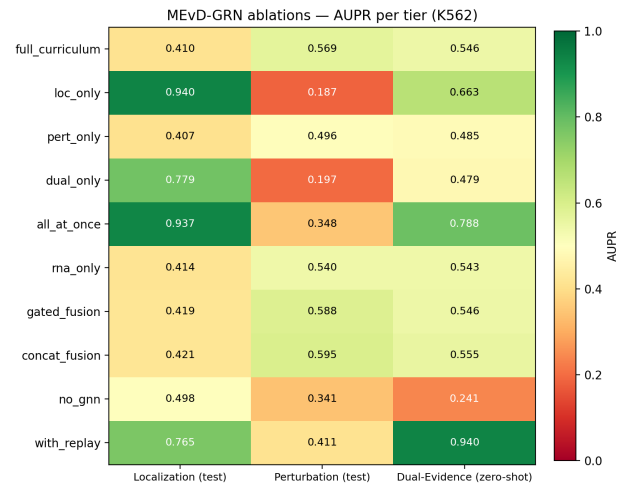


Figure 8. Ablation heatmap: AUPR by configuration $\times$ tier.

A. Full Ablation Metrics

Table 9 reports all four metrics (AUPR, AUROC, EP, EPR) for every ablation configuration and every tier, complementing the AUPR-only view in Table 8.

B. Environment and Reproducibility

Python 3.10, PyTorch ≥ 2.1 , PyTorch Geometric ≥ 2.4 , scikit-learn, pandas, scipy, matplotlib; optional `arboreto` (GRN-Boost2) and `regdiffusion` packages for baselines. All experiments were run on a single consumer GPU (RTX 4060); the full model has under 400K parameters, so a roughly 1:10 GPU:CPU utilization ratio is sufficient. All random splits use a fixed seed (Algorithm 1) for reproducibility.

Table 9. Full ablation results, K562, all metrics. All rows 315,015 params except `with_fm` (3.26M). Single run each (Section 11).

Configuration	Tier	AUPR	AUROC	EP	EPR
<code>full_curriculum</code>	<code>localization</code>	0.568	0.509	0.545	1.0
<code>full_curriculum</code>	<code>perturbation</code>	0.625	0.894	0.576	3.1
<code>full_curriculum</code>	<code>dual_evidence</code>	0.872	0.968	0.784	4.7
<code>loc_only</code>	<code>localization</code>	0.940	0.939	0.870	1.6
<code>loc_only</code>	<code>perturbation</code>	0.185	0.362	0.150	0.8
<code>loc_only</code>	<code>dual_evidence</code>	0.664	0.807	0.649	3.9
<code>pert_only</code>	<code>localization</code>	0.399	0.261	0.378	0.7
<code>pert_only</code>	<code>perturbation</code>	0.487	0.819	0.458	2.5
<code>pert_only</code>	<code>dual_evidence</code>	0.441	0.757	0.370	2.2
<code>dual_only</code>	<code>localization</code>	0.794	0.777	0.727	1.4
<code>dual_only</code>	<code>perturbation</code>	0.189	0.486	0.185	1.0
<code>dual_only</code>	<code>dual_evidence</code>	0.456	0.774	0.524	3.1
<code>all_at_once</code>	<code>localization</code>	0.938	0.929	0.866	1.6
<code>all_at_once</code>	<code>perturbation</code>	0.359	0.747	0.337	1.8
<code>all_at_once</code>	<code>dual_evidence</code>	0.798	0.946	0.707	4.2
<code>rna_only</code>	<code>localization</code>	0.562	0.496	0.529	1.0
<code>rna_only</code>	<code>perturbation</code>	0.583	0.877	0.536	2.9
<code>rna_only</code>	<code>dual_evidence</code>	0.859	0.963	0.771	4.6
<code>coexpr_only</code>	<code>localization</code>	0.530	0.482	0.526	1.0
<code>coexpr_only</code>	<code>perturbation</code>	0.607	0.889	0.563	3.0
<code>coexpr_only</code>	<code>dual_evidence</code>	0.822	0.956	0.739	4.4
<code>tf_cand_only</code>	<code>localization</code>	0.565	0.492	0.515	1.0
<code>tf_cand_only</code>	<code>perturbation</code>	0.590	0.875	0.550	2.9
<code>tf_cand_only</code>	<code>dual_evidence</code>	0.827	0.950	0.735	4.4
<code>gated_fusion</code>	<code>localization</code>	0.572	0.534	0.571	1.1
<code>gated_fusion</code>	<code>perturbation</code>	0.646	0.902	0.592	3.2
<code>gated_fusion</code>	<code>dual_evidence</code>	0.883	0.970	0.795	4.8
<code>concat_fusion</code>	<code>localization</code>	0.579	0.550	0.579	1.1
<code>concat_fusion</code>	<code>perturbation</code>	0.660	0.906	0.603	3.2
<code>concat_fusion</code>	<code>dual_evidence</code>	0.890	0.971	0.798	4.8
<code>edge_mlp</code>	<code>localization</code>	0.578	0.524	0.551	1.0
<code>edge_mlp</code>	<code>perturbation</code>	0.644	0.901	0.592	3.2
<code>edge_mlp</code>	<code>dual_evidence</code>	0.882	0.971	0.795	4.8
<code>gated_relations</code>	<code>localization</code>	0.570	0.512	0.539	1.0
<code>gated_relations</code>	<code>perturbation</code>	0.642	0.900	0.588	3.1
<code>gated_relations</code>	<code>dual_evidence</code>	0.876	0.969	0.791	4.7
<code>no_gnn</code>	<code>localization</code>	0.542	0.531	0.563	1.0
<code>no_gnn</code>	<code>perturbation</code>	0.358	0.721	0.377	2.0
<code>no_gnn</code>	<code>dual_evidence</code>	0.283	0.704	0.299	1.8
<code>with_replay*</code>	<code>localization</code>	0.763	0.719	0.671	1.2
<code>with_replay*</code>	<code>perturbation</code>	0.515	0.824	0.458	2.5
<code>with_replay*</code>	<code>dual_evidence</code>	0.957	0.992	0.890	5.3
<code>with_fm[†]</code>	<code>localization</code>	0.751	0.771	0.736	1.4
<code>with_fm[†]</code>	<code>perturbation</code>	0.834	0.964	0.767	4.1
<code>with_fm[†]</code>	<code>dual_evidence</code>	0.949	0.988	0.869	5.2

*dual-evidence score trains directly on dual-evidence, not zero-shot. [†]3.26M params, not 315,015; see Table 7 for the recommended `all_at_once+FM` configuration’s full metrics.